Alex Ruvolo

W4 Quiz

1.

	Y=2	Y=4	
X=0	0.2	0.3	0.5
X=5	0.4	0.1	0.5
	0.6	0.4	

Not Independent

$Cov(X,Y) = E(XY) - E(X)\cdot E(Y)$

$E(X) = 0\cdot 0.5 + 5\cdot 0.5 = 2.5$

$E(X^2) = 0^2\cdot 0.5 + 5^2\cdot 0.5 = 12.5$

$V(X) = E(X^2) - (E(X))^2$

$V(X) = 12.5 - (2.5)^2 = 6.25$

$SD(X) = \sqrt{6.25} = 2.5$

$E(Y) = 2\cdot 0.6 + 4\cdot 0.4 = 2.8$

$E(Y^2) = 2^2\cdot 0.6 + 4^2\cdot 0.4 = 8.8$

$V(Y) = E(Y^2) - (E(Y))^2$

$V(Y) = 8.8 - (2.8)^2 = 0.96$

$SD(Y) = \sqrt{0.96} = 0.98$

$E(XY) = 0\cdot 2\cdot 0.2 + 0\cdot 4\cdot 0.3 + 5\cdot 2\cdot 0.4 + 5\cdot 4\cdot 0.1$

$= 0 + 0 + 4 + 2 = 6$

$Cov(X,Y) = 6 - 2.5\cdot 2.8 = -1$

$$\boxed{Corr(X,Y) = \frac{-1}{2.5\cdot 0.98} = -0.41}$$

2. $Y = X_1 + X_2 + X_3 + X_4 + X_5$

$M_Y(t) = e^{15e^t - 15}$

Poisson $\rightarrow M(t) = e^{\lambda(e^t - 1)} \rightarrow \lambda = 15$

$Y \sim Poi(15)$

$X_1, X_2, X_3, X_4, X_5 \sim Poi(3)$

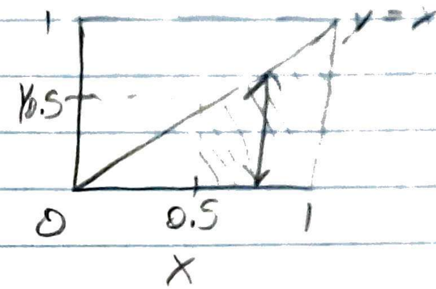
$\boxed{V(X_1) = 3}$

W4 Quiz

3. $f(x,y) = k(x+y),\ 0 < y < x < 1$ Not Independent

$6\,m \to 0.5$ years

$P(x > 0.5 \cup y > 0.5$



$P(x > 0.5 \cup y > 0.5)$

$= k \int_{0.5}^{1} \int_{0}^{x} x + y \, dy \, dx$

$= k \int_{0.5}^{1} xy + \frac{1}{2}y^2 \Big|_0^x \, dx$

$= k \int_{0.5}^{1} x^2 + \frac{1}{2}x^2 \, dx$

$= k \int_{0.5}^{1} \frac{3}{2} x^2 \, dx$

$= k(\frac{1}{2}x^3)\Big|_{0.5}^{1}$

$= k((\frac{1}{2}(1)^3) - (\frac{1}{2}(0.5)^3))$

$= k(\frac{1}{2} - \frac{1}{16})$

$= k(\frac{8}{16} - \frac{1}{16})$

$= \boxed{\frac{7}{16}k}$